

Problem Set 9

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1 Variance and Covariance

Problem 1: Suppose I draw two consecutive cards from a five card deck numbered 2 through 6. Let X be the random variable corresponding to the first card and Y corresponding to the second card, where each maps the card to their face value.

1. What is the variance of X ?
2. What is the covariance of X and Y ?

Problem 2: Let P be a random variable corresponding to value on the throw of a 20-sided die (with sides 1 through 20). Find a random variable Q such that the correlation between P and Q is:

1. 1
2. 0
3. -1

Show your work.

2 Markov and Chebyshev

Problem 3: Let's slightly modify the game of Blackjack such that Ace is worth one point and all other cards remain the same. (Number cards are worth their face value and Jack, Queen and King are worth 10).

1. Use Markov's Inequality to bound the probability of getting 19 or more points upon drawing two cards from a standard deck.
2. What is the actual probability of drawing two cards worth 19 or higher?

Problem 4: Suppose you roll 4 dice. Bound the probability of obtaining a total greater than 20 using

- Markov's Inequality
- Chebyshev's Inequality